


# U23CC583 – DIGITAL AND MOBILE FORENSICS LAB MANUAL

Leadership & Excellence

# Sri Eshwar College of Engineering

Coimbatore | Tamilnadu

Kondampatti (Post), Vadasithur (Via), Coimbatore - 641 202.

*Department of*

.....

*Record work of ..... Laboratory*

*Certified bonafide Record of work done by*

*Name : ..... Roll No. : .....*

*Class : ..... Branch : .....*

*Reg. No. : .....*

*Place :* *HOD* *Faculty In charge*

*Date :*

*University Register No. .....*

*Submitted for the University Practical Examination held on*

.....

*Internal Examiner* *External Examiner*

# Instructions for LABORATORY Classes

- Enter the Lab with observation, record notebook, calculator & necessary tools
- Enter the Lab with **CLOSED FOOTWEAR**
- Maintain Silence during the Lab Hours
- Boys should **"TUCK IN"** the Shirts
- Girls are instructed to wear **OVERCOATS**
- H**ANGING CHAINS, RINGS, WRIST WATCHES** and like items are likely to cause accidents and hence are to be **AVOIDED**
- **LONG HAIR** should be protected, let it not be loose especially near **ROTATING MACHINERY**
- Any machine / equipment **SHOULD NOT BE OPERATED** other than the prescribed one for the day
- Read and follow the work instructions inside the laboratory
- **POWER SUPPLY** to your table should be obtained only through the **LAB TECHNICIAN**
- Do not **LEAN** and do not be **CLOSE** to the rotating components
- Handle the equipments with care
- **TOOLS & GAUGE** sets are to be **RETURNED** before leaving the lab
- **ALL SUB - HEADING** and **DETAILS** should be neatly written on the right hand side page of the record
- For example

  Aim of the experiment

  Apparatus / Tools / Instruments required

  Procedure

  Model calculation

  Results / discussions / inference
- The sketches, block diagrams, circuits and flow charts are to be drawn on the left side blank page
- For example

  Neat Diagram

  Specifications / Design Details

  Tabulations

  Graph
- **GRAPHS / FIGURES** drawn on separate sheets in special cases, should be firmly pasted on to the record.
- Before writing the record, the student should get the corresponding observations signed by the **FACULTY - IN- CHARGE**
- The observation and record should be completed in all respects and submitted in the very **NEXT CLASS** itself.
- Begin every experiment on a **NEW PAGE**
- Write the **NAME OF EXPERIMENT** in Capital letters on the top of the page
- Experiment number with date should be written at the top left hand corner
- Be **PATIENT, STEADY, SYSTEMATIC** and **REGULAR**
- Follow the dress code for Laboratory classes
- Maintain punctuality

## Record Index

| S. No. | Date | Name of the Experiment | Page No. | Algorithm (20) | Procedure (25) | Inference (20) | Viva (10) | Total (75) | Faculty Sign |
|---|---|---|---|---|---|---|---|---|---|
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Signature of the Faculty member

## VISION & MISSION OF THE DEPARTMENT

**Vision:**

"To emerge as a pioneering department, nurturing students into futuristic and globally competent Computer and Communication Engineering professionals."

**Mission:**

M1: To offer high-quality engineering education emphasizing practical outcomes.

M2: To foster innovative research and project development among students and instil a sense of social responsibility.

M3: To equip students with industry-relevant skills and robust leadership qualities through pedagogical methodologies and industry engagement.

M4: To inculcate a culture of socially relevant values and nurture moral, ethical and interpersonal skills among students.

**Program Educational Objectives (PEOs):**

**PEO1:** Acquire core competencies in computational technologies communication engineering with the ability to utilise the latest software tools to effectively address real-time challenges.

**PEO2:** Active involvement in research, innovation, or entrepreneurial ventures, resulting in significant societal contributions.

**PEO3:** Demonstrate industry-relevant skills, effective leadership qualities, and strong ethical values to exhibit professionalism and foster collaborative work experience.

**Program Outcomes:**

| **PROGRAM OUTCOMES (POs)** | | **Graduates of Computer and Communication Engineering will be able to** |
|---|---|---|
| **PO1.** | **Engineering knowledge** | Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization to develop the solution of complex engineering problems. |
| **PO2.** | **Problem analysis** | Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. |
| **PO3.** | **Design/developmentof solutions** | Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health and safety, whole-life cost, net zero carbon, culture, society, and environment as required. |

| | | |
|---|---|---|
| **PO4.** | **Conduct investigations of complex problems** | Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. |
| **PO5.** | **Engineering Tool usage** | Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. |
| **PO6.** | **The Engineer and The World** | Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture, and environment. |
| **PO7.** | **Ethics** | Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. |
| **PO8.** | **Individual and Collaborative Teamwork** | Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams. |
| **PO9.** | **Communication** | Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences. |
| **PO10.** | **Project Management and Finance** | Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments. |
| **PO11.** | **Life-Long Learning** | Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. |

**Program Specific Outcomes (PSO):**

**PSO1:** Apply the concepts of computer and communication systems to develop software for different applications.

**PSO2:** Acquire skills and knowledge to solve real-time challenges in communication networks.

**Course Outcomes COs:**

| CO. No. | Course Outcome | BTL | POs | PSOs |
|---|---|---|---|---|
| U23CC583.1 | Analyze the digital forensics process to investigate cyber incidents. | K4 | 1,2,3,4,5,6,7 | 1 |
| U23CC583.2 | Apply digital crime investigation techniques and international cooperation methods to collect and analyze digital evidence. | K3 | 1,2,3,4,5,6,7 | 1 |
| U23CC583.3 | Analyze digital forensic readiness framework and methodologies to enhance forensic capabilities in enterprises. | K4 | 1,2,3,4,5,6,7 | 1 |
| U23CC583.4 | Apply iOS forensic techniques to analyze mobile device data. | K3 | 1,2,3,4,5,6,7 | 1 |
| U23CC583.5 | Apply Android forensic techniques to analyze mobile device data. | K3 | 1,2,3,4,5,6,7 | 1 |

**CO & PO/PSO Mapping:**

| CO No. | Programme Outcomes | | | | | | | | | | | | PSO | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | PO-01 | PO-02 | PO-03 | PO-04 | PO-05 | PO-06 | PO-07 | PO-08 | PO-09 | PO-10 | PO-11 | PO-12 | PSO-01 | PSO-02 |
| R19CC352.1 | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |
| R19CC352.2 | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |
| R19CC352.3 | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |
| R19CC352.4 | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |
| R19CC352.5 | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |
| Average | 3 | 2 | 1 | 1 | 3 | 1 | 1 | - | - | - | - | - | 2 | - |

"3"—High, "2"—Medium, "1"—Low, "-"—No Correlation

Syllabus:

R19CC352 – DIGITAL AND MOBILE FORENSICS

| List of Experiments: (30 periods) |
|---|
| 1. Implementation of Metadata Extraction from a Digital Image. (CO1) |
| 2. Recovery and Analysis of Deleted Files from a USB Drive. (CO2) |
| 3. Implementation of Log Monitoring for Forensic Readiness. (CO3) |
| 4. Analysis of Data Extraction from an iOS Backup. (CO4) |
| 5. Implementation of Data Extraction from an Android Device using ADB. (CO5) |

## Ex.No : 01 IMPLEMENTATION OF METADATA EXTRACTION FROM A DIGITAL IMAGE

**Date :**

### AIM:

To extract, examine, and analyze metadata from a digital image using a forensic tool and interpret its significance in digital forensic investigations.

### SOFTWARE/EQUIPMENTS REQUIRED:

- Windows 10/11 or Linux (Ubuntu/Kali)
- ExifTool

### THEORY:

Metadata is commonly referred to as "data about data." It is descriptive information stored within or associated with a digital file that provides details about the file's properties, origin, creation, and modification history. Although metadata is typically invisible during normal file viewing, it can be extracted using specialized forensic tools. In digital forensics, metadata plays a crucial role in investigating digital evidence. It helps investigators determine the origin, authenticity, and history of a file, reconstruct event timelines, verify evidence integrity, and identify possible signs of tampering or manipulation.

### TOOL USED:

ExifTool is a command-line tool used to read, write, edit, and extract metadata from digital files such as images, videos, audio, and documents. It is widely used in digital forensics to analyze file metadata.

### ALGORITHM:

1. Download and install ExifTool.
2. Open Command Prompt (Windows) or Terminal (Linux).
3. Navigate to the folder containing the image.
4. Execute the following command to extract the metadata.

   Example: exiftool image1.jpg
5. Observe the metadata displayed by ExifTool.
6. Record important metadata fields.
   - File Name, File Size
   - File Type
   - MIME Type
   - Image Width
   - Image Height
   - Date Created, Date Modified
   - Camera Make, Camera Model
7. Interpret the extracted metadata and document your observations.
8. End

## PROCEDURE

### Step 1: Download ExifTool

1. Dowload ExifTool by Phil Harvey.
2. Find the Windows Executable download.
3. Download the Windows ZIP file.
4. Extract the downloaded ZIP file.
5. Locate the executable file, usually named exiftool(-k).exe.
6. Rename exiftool(-k).exe to exiftool.exe.
7. Keep the executable in a simple folder, such as D:\DMF\ExifTool.

### Step 2: Prepare the experiment folder

1. Open File Explorer.
2. Create the folder D:\DMF\Experiment1.
3. Copy one sample JPEG image into this folder.
4. Rename the image to image1.jpg.
5. If the extracted package contains an exiftool_files folder, keep that folder alongside exiftool.exe.

### Step 3: Open Command Prompt

1. Press Windows + R.
2. Type cmd.
3. Press Enter.

### Step 4: Navigate to the experiment folder

1. Type this command and press Enter:

```bat
cd /d D:\DMF\Experiment1
```

### Step 5: Extract the image metadata

```bat
exiftool.exe image1.jpg
```

**Sample Output:**

```
File Name                       : image1.jpg
Directory                       : D:/DMF/Experiment1
File Size                       : 245 kB
File Type                       : JPEG
MIME Type                       : image/jpeg
Image Width                     : 1920
Image Height                    : 1080
Image Size                      : 1920x1080
```

**Output**

| S.No. | Item | Description |
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## Ex.No : 02 RECOVERY AND ANALYSIS OF DELETED FILES FROM A USB DRIVE

**Date :**

**AIM:**

To recover deleted files from a USB drive using a digital forensic tool and analyze the recovered data for forensic investigation.

**SOFTWARE/EQUIPMENTS REQUIRED:**

- Windows 10/11 or Linux (Ubuntu/Kali)
- TestDisk , PhotoRec

**THEORY:**

Deleted files are not immediately removed from a storage device. When a file is deleted, the operating system typically marks its storage space as available for reuse, while the actual data remains on the device until it is overwritten. Digital forensic tools can recover these files by scanning the file system or performing signature-based recovery. File recovery is an essential process in digital forensics, cybercrime investigations, and incident response.

**TOOL USED:**

PhotoRec is an open-source command-line data recovery tool used to recover deleted files from storage devices such as USB drives, hard disks, memory cards, and SSDs. It uses file signature-based recovery and supports a wide range of file systems, including FAT, NTFS, exFAT, ext2/ext3/ext4, HFS+, and others. It is widely used in digital forensics to recover deleted or lost files from damaged or formatted storage media.

**ALGORITHM:**

1. Start the experiment and ensure that TestDisk/PhotoRec is installed on the system.
2. Connect the test USB drive and identify its device name and storage capacity.
3. Launch PhotoRec with the required administrative privileges.
4. Select the USB drive from the list of available storage devices.
5. Select the appropriate partition and filesystem type.
6. Choose the Free option to scan unallocated space for deleted files, or Whole to scan the entire partition.
7. Select a destination directory on a different storage device to store the recovered files.
8. Start the recovery process and wait for PhotoRec to complete the scan.
9. Examine the recovered files and record the file types, recovery count, and recovery status.
10. Document the observations and forensic findings.
11. End

## PROCEDURE

### Step 1: Download PhotoRec

1. Download the Windows version of TestDisk.
2. Extract the downloaded ZIP file to a folder such as C:\DMF\Exp 2\TestDisk.
3. Open the extracted folder.
4. Locate photorec_win.exe.

### Step 2: Prepare a test USB drive

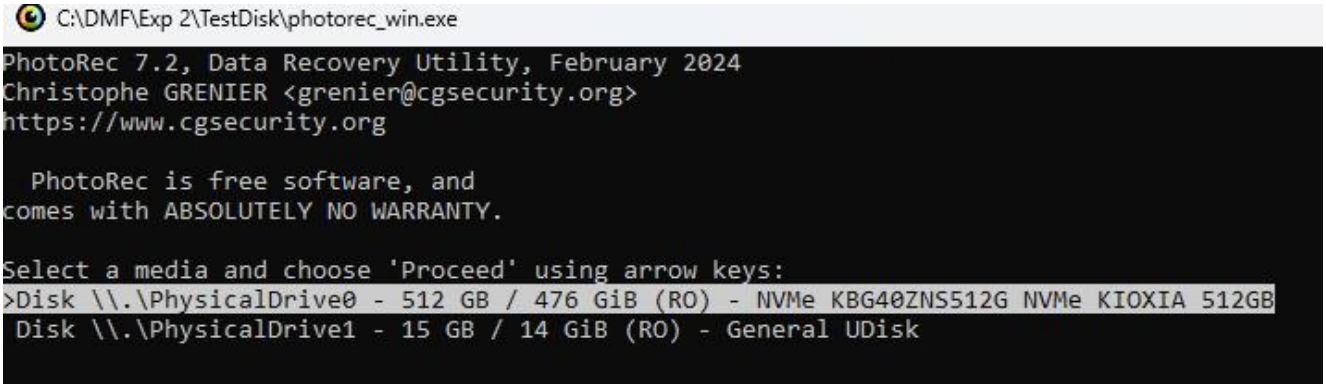
1. If you do not already have deleted sample files, create a practice dataset first.
2. Connect a USB drive to your computer.
3. Create a folder named DMF_Test.
4. Copy a few sample files into it, such as: sample.txt, sample.jpg, sample.pdf
5. Open each file once to confirm it works.
6. Delete these test files and empty the Recycle Bin if prompted.
7. Do not copy new files onto the USB drive after deleting them. Recovery may fail if the deleted data has been overwritten.

### Step 3: Launch PhotoRec

1. Open the extracted TestDisk folder.
2. Right-click photorec_win.exe.
3. Select Run as administrator.
4. Approve the Windows permission prompt.

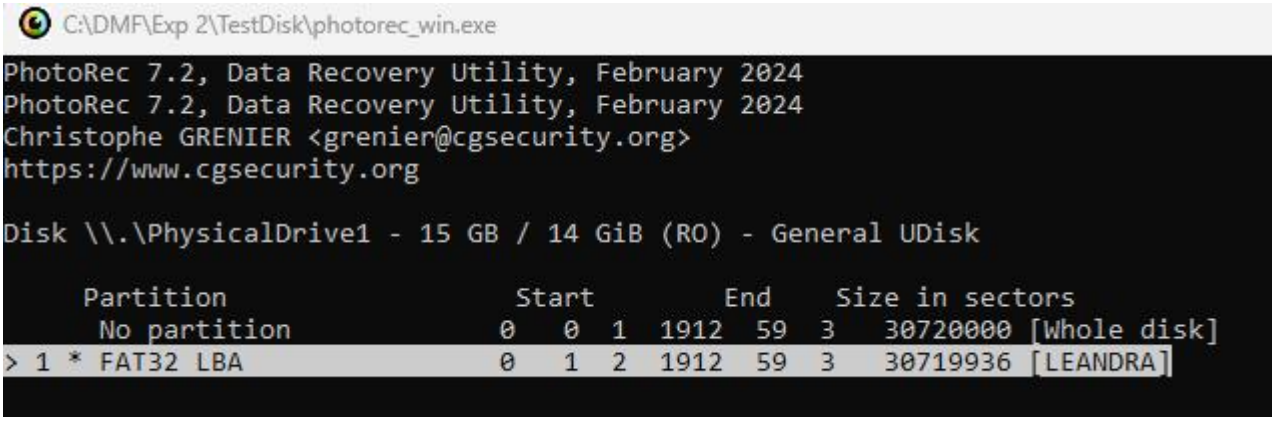
### Step 4: Select the USB drive

1. Use the Up/Down arrow keys to select your USB drive.
2. Identify it by its storage capacity. For example, a 16 GB USB drive should be distinguishable from your computer's internal drive.
3. Press Enter to proceed.



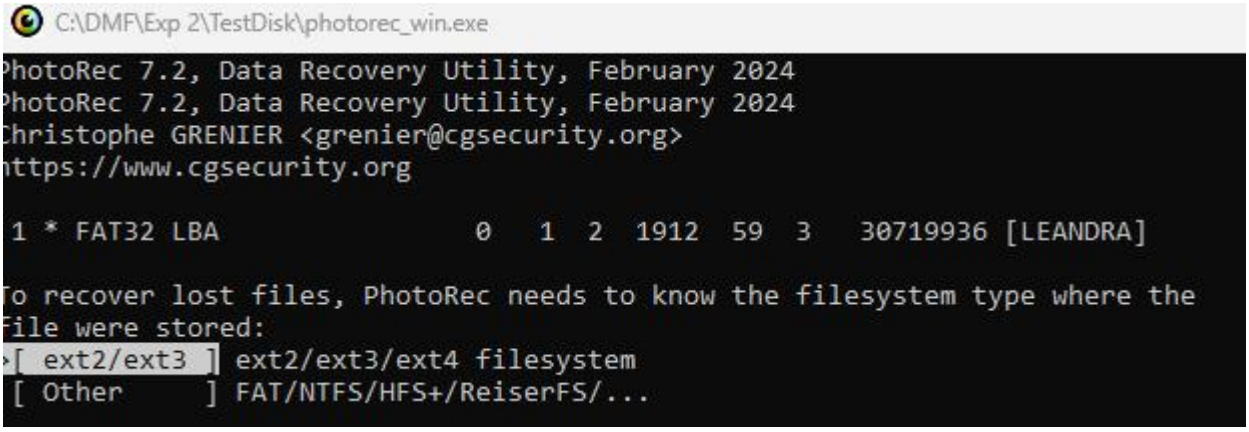

### Step 5: Select the partition

1. Select the partition belonging to your USB drive.
2. Choose Search and press Enter.
3. If the partition is not shown as expected, stop and check that you selected the correct device.



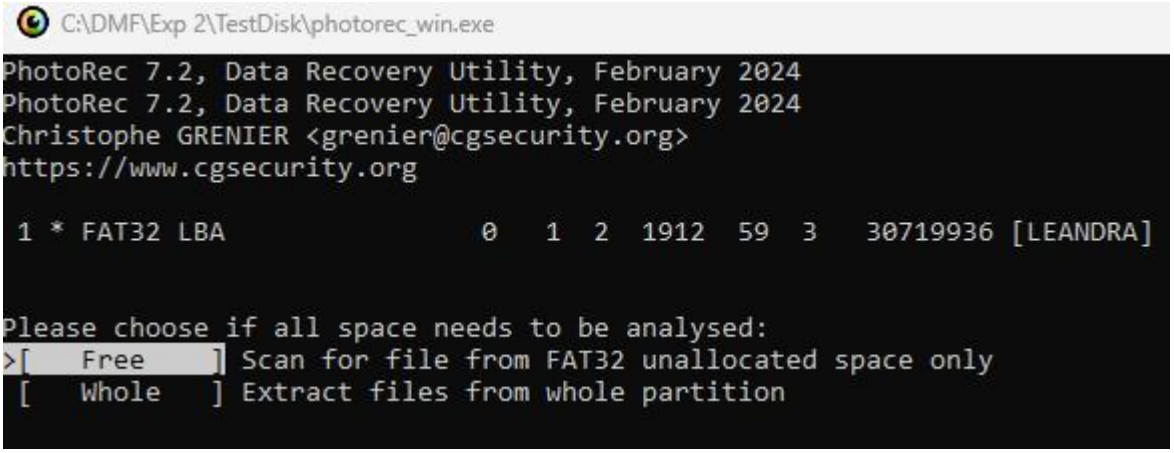

**Step 6: Select the filesystem type**

1. PhotoRec will ask which filesystem family was used.
2. Select Other for common Windows USB filesystems such as FAT32, exFAT, and NTFS.
3. Select the ext2/ext3/ext4 option only if the partition uses a Linux filesystem.




**Step 7: Choose the scan area**

1. PhotoRec will offer scanning choices.
2. Free — scans unallocated space for deleted files, where supported.
3. Whole — scans the entire selected partition.

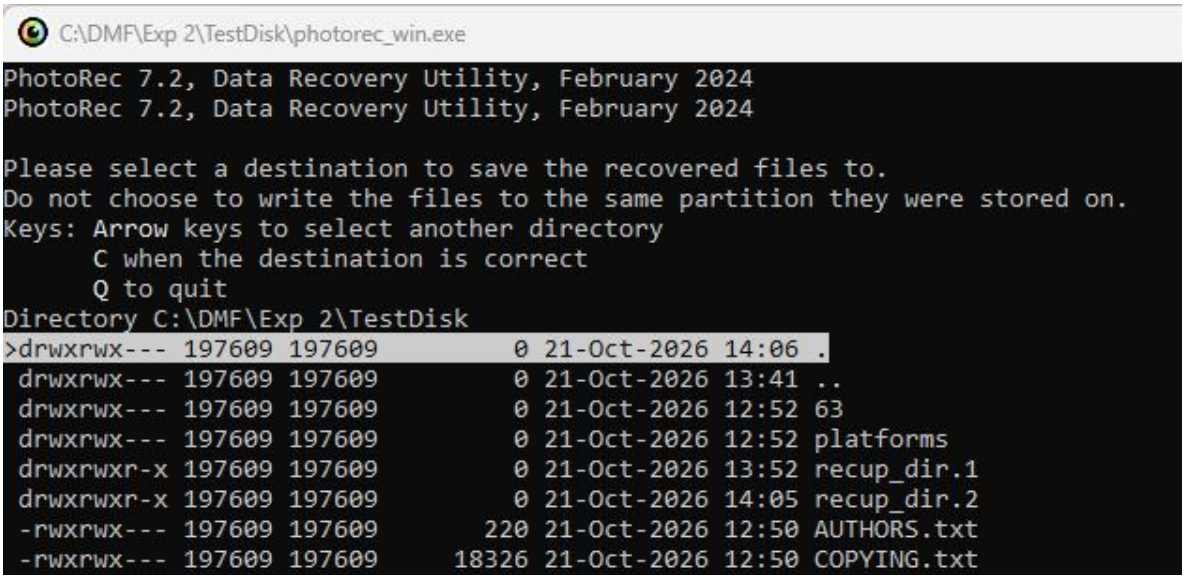



**Step 8: Choose the recovery destination**

1. Create a folder on your computer, for example C:\DMF\Recovered.
2. In PhotoRec, use the arrow keys to navigate to the destination folder.
3. Press Enter to enter folders.

4. When the correct destination is selected, press the key indicated by PhotoRec to confirm it (commonly C in the documented interface).

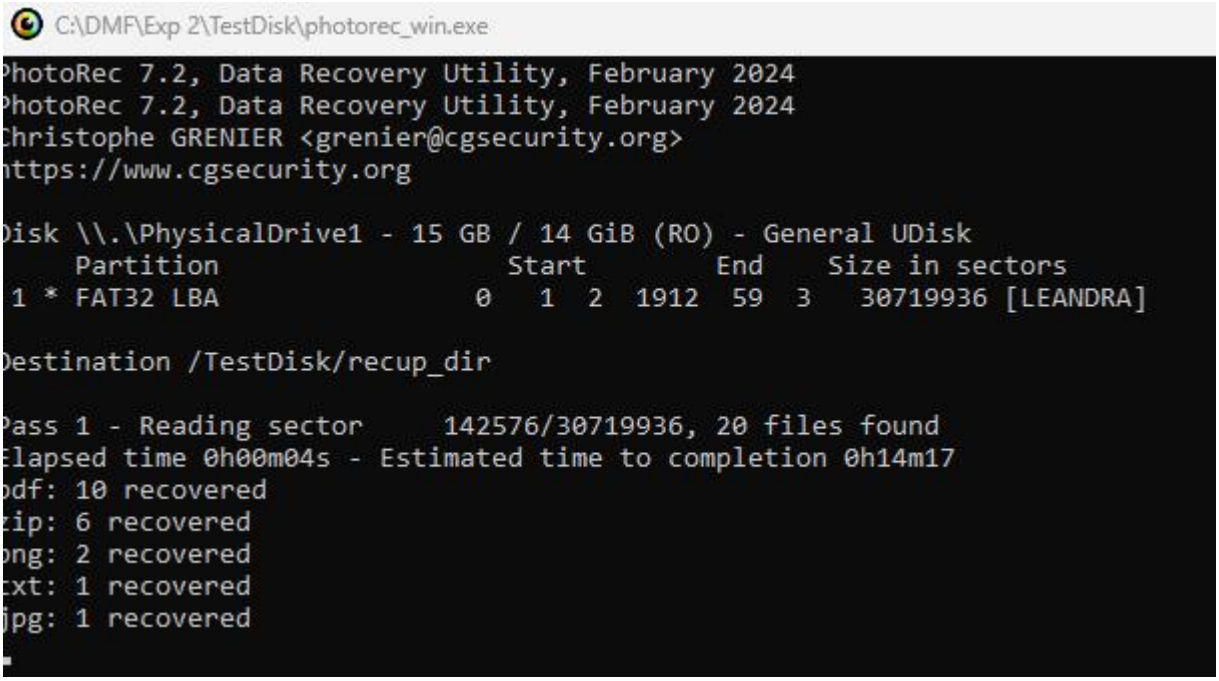
5. Never select the USB drive being scanned as the destination. Recovered files must be written to a different drive or filesystem.




**Step 9: Recover the files**

1. PhotoRec starts scanning the selected area, Wait for the scan to complete.
2. Observe the recovery count displayed on the screen.
3. After completion, exit PhotoRec.
4. The recovered files are normally saved in folders named recup_dir.1, recup_dir.2, and so on, inside the destination folder. The original filenames and folder structure may not be preserved.

## Output:



## Ex.No: 03 IMPLEMENTATION OF LOG MONITORING FOR FORENSIC READINESS

**Date:**

**AIM:**

To implement log monitoring for forensic readiness.

**SOFTWARE/EQUIPMENTS REQUIRED:**

- Windows 10/11
- Windows Powershell

**THEORY:**

Log monitoring is an important component of digital forensic readiness because system logs contain records of activities performed on a computer. These records may include login attempts, authentication failures, system activities, timestamps, and other security-related events. Windows maintains security-related events in the Windows Security Event Log. By continuously monitoring these events, suspicious activities can be identified and recorded for further investigation.

**TOOL USED:**

PowerShell is used to access and monitor the Windows Security log. Event ID 4625 is used to identify failed login attempts. Multiple failed login attempts are treated as suspicious activity for the purpose of this experiment. The identified events are then exported to a CSV file so that they can be preserved and analyzed as potential forensic evidence.

**ALGORITHM:**

1. Start the program.
2. Open Windows PowerShell with administrator privileges.
3. Access the Windows Security Event Log.
4. Generate a few failed login attempts on the local computer.
5. Search the Security log for Event ID 4625.
6. Retrieve and count the detected failed login events.
7. If the count is three or more, display a suspicious activity alert.
8. Create a folder to preserve forensic evidence.
9. Export the detected security events to a CSV file.
10. Verify the exported file and examine its contents.
11. Preserve the evidence.
12. End

## PROCEDURE:

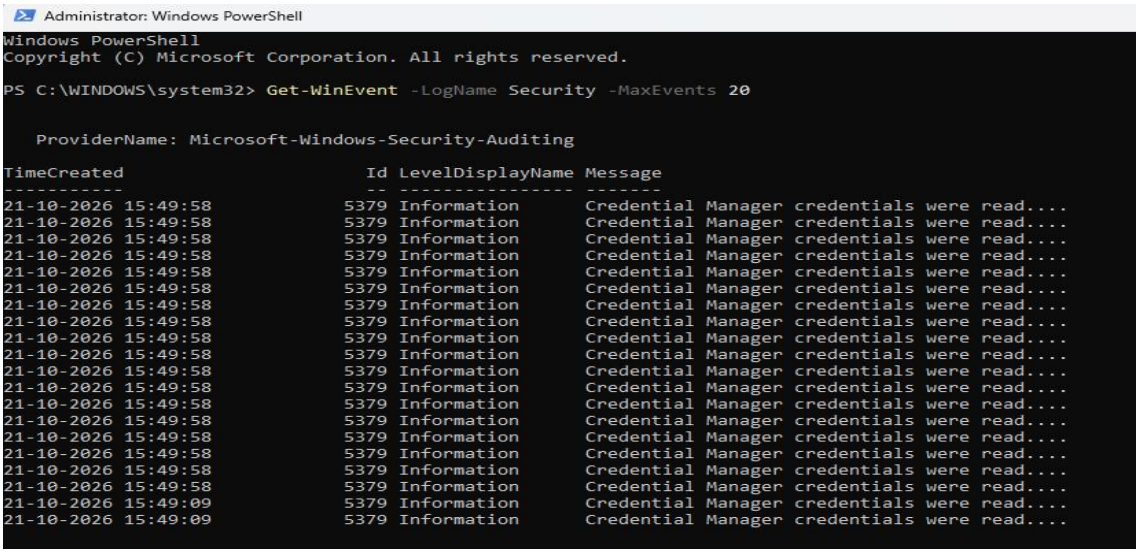
### Step 1: Open PowerShell as Administrator

1. Click the Windows Start button.
2. Search for `PowerShell`.
3. Right-click Windows PowerShell.
4. Select Run as administrator.

### Step 2: Check the Security Log

1. Copy and run this command:

Get-WinEvent -LogName Security -MaxEvents 20



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> Get-WinEvent -LogName Security -MaxEvents 20

   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
21-10-2026 15:49:58           5379 Information      Credential Manager credentials were read....
21-10-2026 15:49:58           5379 Information      Credential Manager credentials were read....
21-10-2026 15:49:58           5379 Information      Credential Manager credentials were read....
21-10-2026 15:49:58           5379 Information      Credential Manager credentials were read....
21-10-2026 15:49:58           5379 Information      Credential Manager credentials were read....
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21-10-2026 15:49:09           5379 Information      Credential Manager credentials were read....
21-10-2026 15:49:09           5379 Information      Credential Manager credentials were read....
```

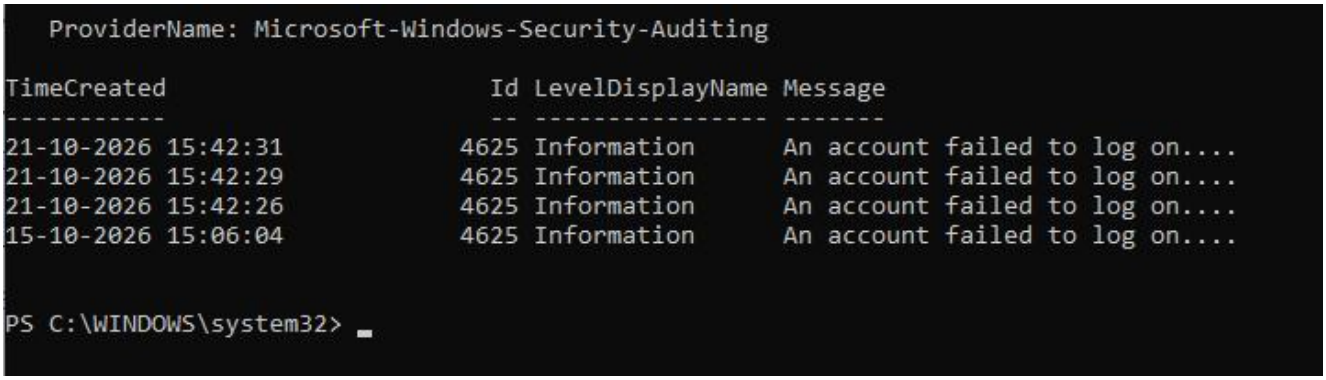
### Step 3: Generate failed login attempts

1. Press Windows + L to lock the computer.
2. Enter an incorrect password a few times.
3. Enter the correct password to unlock the computer.

### Step 4: Find failed login events

Return to PowerShell and run:

```
Get-WinEvent -FilterHashtable @{
    LogName = 'Security'
    Id = 4625
} -MaxEvents 20
```



```
   ProviderName: Microsoft-Windows-Security-Auditing

TimeCreated                     Id LevelDisplayName Message
-----------                     -- ---------------- -------
21-10-2026 15:42:31           4625 Information      An account failed to log on....
21-10-2026 15:42:29           4625 Information      An account failed to log on....
21-10-2026 15:42:26           4625 Information      An account failed to log on....
15-10-2026 15:06:04           4625 Information      An account failed to log on....


PS C:\WINDOWS\system32>
```

**Step 5: Count failed login events and generate an alert**

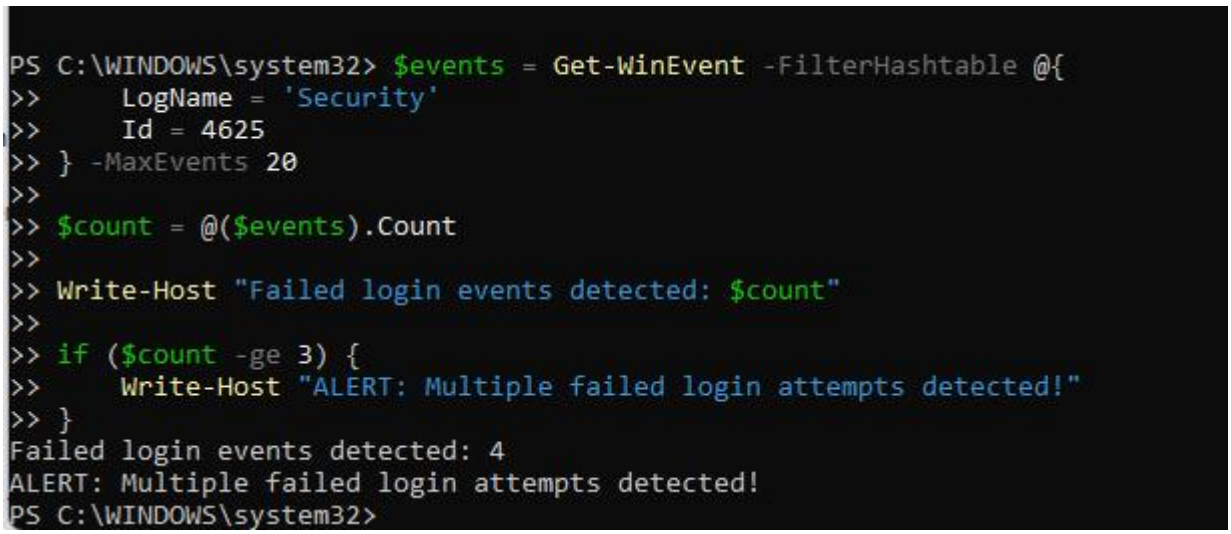
Copy and run the complete block:

```
$events = Get-WinEvent -FilterHashtable @{
    LogName = 'Security'
    Id = 4625
} -MaxEvents 20

$count = @($events).Count

Write-Host "Failed login events detected: $count"

if ($count -ge 3) {
    Write-Host "ALERT: Multiple failed login attempts detected!"
}
```




**Step 6: Create a folder for forensic evidence**

We will use a simple folder path on the C drive.

Run:

New-Item -ItemType Directory -Path "C:\DMF\Exp3" -Force

**Step 7: Export the failed login events to CSV**

Run the following command:

Get-WinEvent -FilterHashtable @{

LogName = 'Security'

Id = 4625

} -MaxEvents 20 |

Select-Object TimeCreated, Id, ProviderName, MachineName, Message |

Export-Csv "C:\DMF\Exp3\forensic_events.csv" -NoTypeInformation

The file will be saved here:

`C:\DMF\Exp3\forensic_events.csv`

This file contains the timestamps, event IDs, computer information, and event messages.

# OUTPUT

| TimeCreated | Id | ProviderName | MachineName | Message |
|---|---|---|---|---|
| 21-10-2026 15:42 | 4625 | Microsoft-Windows-Security-Auditing | SECE | An account failed to log on.<br><br>Subject:<br>Security ID:S-1-5-18<br>Account Name:SECE$<br>Account Domain:WORKGROUP<br>Logon ID:0x3E7<br><br>Logon Type:2<br><br>Account For Which Logon Failed:<br>Security ID:S-1-0-0<br>Account Name:-<br>Account Domain:-<br><br>Failure Information:<br>Failure Reason:An Error occured during Logon.<br>Status:0xC000006D<br>Sub Status:0xC0000380<br><br>Process Information:<br>Caller Process ID:0x8a4<br>Caller Process Name:C:\Windows\System32\svchost.exe<br><br>Network Information:<br>Workstation Name:-<br>Source Network Address:127.0.0.1 |

## Ex.No : 04 ANALYSIS OF DATA EXTRACTION FROM AN IOS BACKUP

**Date :**

## AIM:

To analyze an iOS device backup and extract forensic information such as contacts, messages, call history, application data, and metadata without requiring a physical iPhone.

## SOFTWARE/EQUIPMENTS REQUIRED:

**Hardware**

- Computer/Laptop
- Digital image file (JPG/JPEG captured using a smartphone or digital camera)

**Software**

- Windows or Linux Operating System
- iLEAPP (iOS Logs, Events, And Plists Parser) for extracting and analyzing iOS backup artifacts
- Python 3 to run iLEAPP.
- Command Prompt or PowerShell.
- A sample iOS backup or authorized forensic dataset.

## THEORY:

An iOS backup contains data from an iPhone or iPad, including application data, messages, contacts, call history, settings, and other device artifacts, depending on the backup type and configuration. In digital forensics, analyzing an iOS backup helps investigators recover and interpret relevant digital evidence. iOS backups may contain databases (SQLite), property list files (PLIST), application files, and media such as photos and videos. Tools such as iLEAPP (iOS Logs, Events, And Plists Parser) help extract and analyze supported artifacts from iOS backups and filesystem extractions.

In digital forensics, image metadata helps an investigator to:

- Recover digital artifacts: Examine available messages, contacts, call records, browsing activity, and application data.
- Establish a timeline: Analyze timestamps and records to understand the sequence of events.
- Identify application activity: Examine supported application databases, logs, and configuration files.
- Support investigations: Correlate multiple artifacts to understand relevant user or device activity.

To preserve evidence integrity, investigators should retain the original iOS backup, analyze a verified copy, record and verify its SHA-256 hash, and document all acquisition and analysis procedures

## ALGORITHM:

1. Download and install ExifTool.
2. Start the Android emulator using Android Studio.
3. Verify the ADB installation and emulator connection.
4. Retrieve the Android device information using ADB commands.
5. Create a sample text file on the computer.
6. Transfer the sample file to the Android emulator using the `adb push` command.
7. Verify the presence of the file on the emulator.
8. Extract the file from the emulator to the computer using the `adb pull` command.
9. Calculate the SHA-256 hashes of the original and extracted files.
10. Compare both hash values to verify file integrity.
11. End

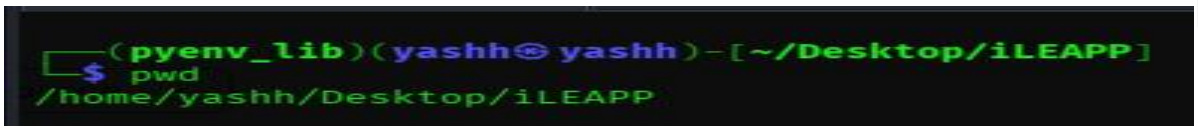
## PROCEDURE

**Step 1: Install iLEAPP**

- Download iLEAPP from the official GitHub repository.
- Install Python 3 and the required dependencies.
- Extract the iLEAPP ZIP file into a working folder, for example C:\iOSForensics\iLEAPP.

Verify the installation by opening Command Prompt in that folder and running:

```
cd iLEAPP
```

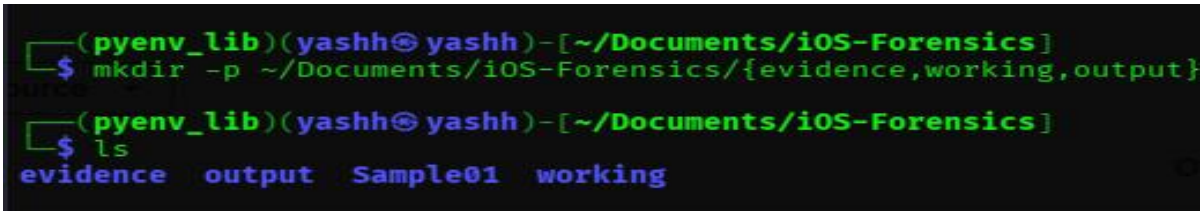



**Step 2:** Prepare the iOS Backup

- Obtain an authorized iOS backup or public forensic sample dataset.
- Copy the backup into C:\iOSForensics\evidence without modifying the original.
- Create a separate working copy for analysis.

**Step 3:** Create the Evidence and Output Folders

Create separate folders to preserve the original dataset, analyze a working copy, and store the results.




**Step 4: Launch the iLEAPP Application**

- Open the terminal and navigate to the iLEAPP directory.
- Activate the Python virtual environment if required.
- Run the following commands to launch the application:

```
┌──(pyenv_lib)(yashh㉿yashh)-[~/Desktop/iLEAPP]
└─$ python ileapp
ileappGUI.py  ileapp.py
┌──(pyenv_lib)(yashh㉿yashh)-[~/Desktop/iLEAPP]
└─$ python ileappGUI.py
```

- The iLEAPP graphical user interface will open. Click the input selection button and select the downloaded iOS forensic image or supported backup.
- Select the appropriate input type and specify the output folder as ~/Documents/iOS-Forensics/output. Click the option to start processing and wait for the analysis to finish.
- Once processing is complete, open the generated HTML report from the output folder and examine the extracted forensic artifacts.

**Step 5: Analyze the Extracted Data**

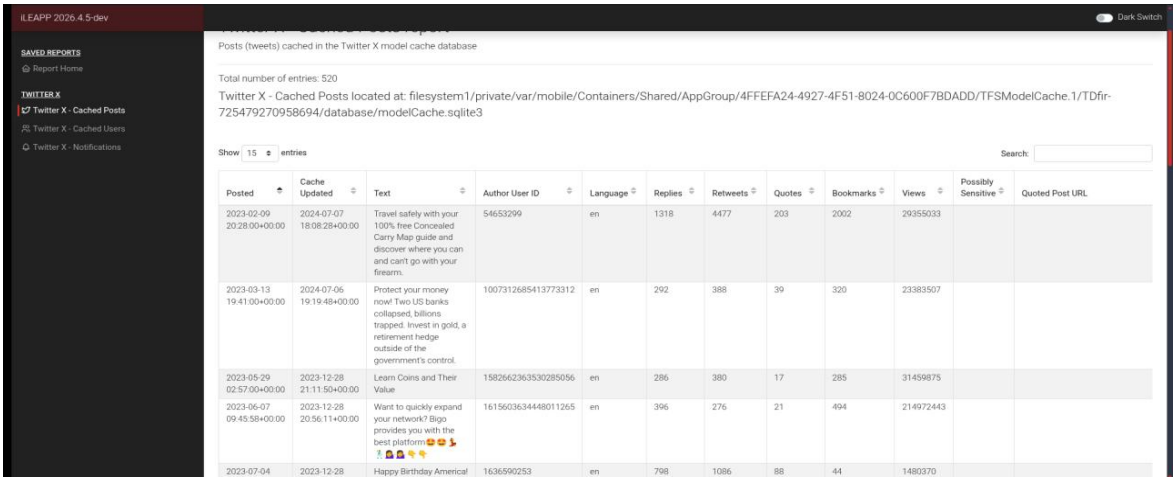
1. After processing is complete, open the output folder selected in the iLEAPP graphical user interface.
2. Open the generated HTML report in a web browser.
3. Examine the extracted forensic artifacts, including messages, contacts, application data, logs, and timestamps, where available.
4. Review the relevant records and identify information useful for the investigation.
5. Record the findings, including artifact details and timestamps, in the forensic report.

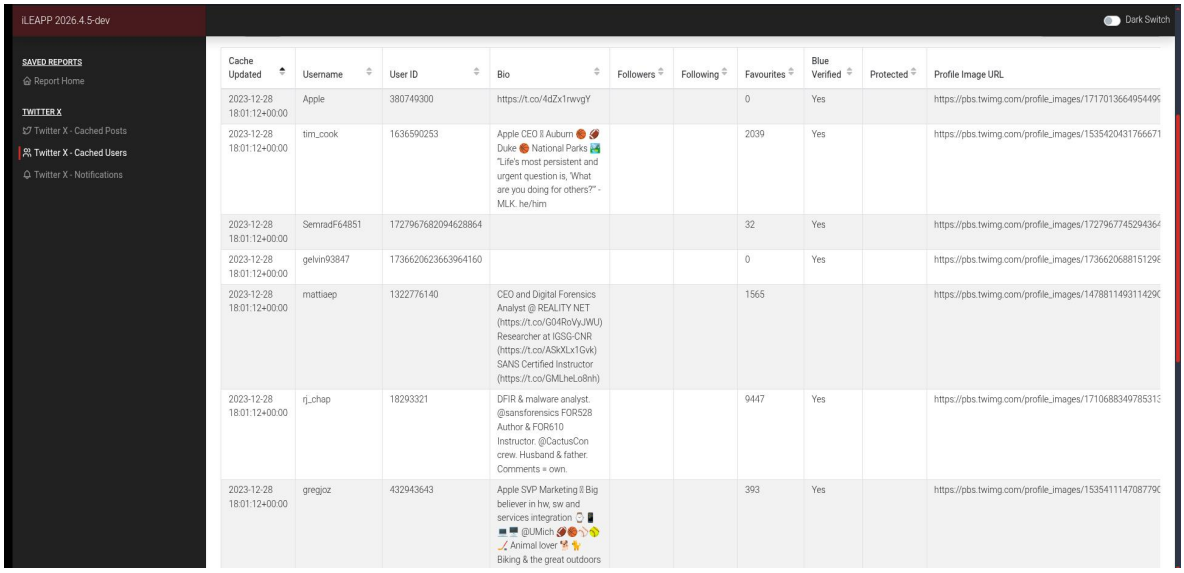
```
┌──(pyenv_lib)(yashh㉿yashh)-[~/Desktop/iLEAPP]
└─$ ls ~/Documents/iOS-Forensics/output
 data     index.html            _lava_data.lava    _Timeline
 _HTML    _lava_artifacts.db    media              '_TSV Exports'
```

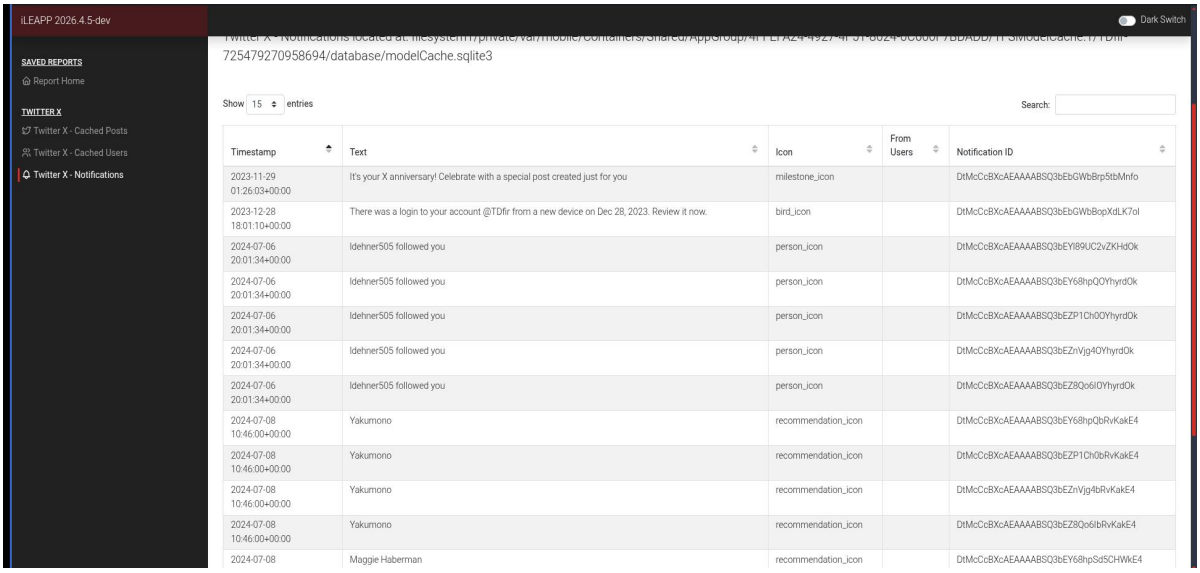
**Step 6:Verify Evidence Integrity and View the Report**

After analysis, calculate the hash again.

# OUTPUT:


iLEAPP 2026.4.5-dev
Dark Switch
SAVED REPORTS
Report Home
TWITTER X
Twitter X - Cached Posts
Twitter X - Cached Users
Twitter X - Notifications
Posts (tweets) cached in the Twitter X model cache database
Total number of entries: 520
Twitter X - Cached Posts located at: filesystem1/private/var/mobile/Containers/Shared/AppGroup/4FFEFA24-4927-4F51-8024-0C600F7BDADD/TFSModelCache.1/TDfir-725479270958694/database/modelCache.sqlite3
Show 15 entries
Search:
Posted
Cache Updated
Text
Author User ID
Language
Replies
Retweets
Quotes
Bookmarks
Views
Possibly Sensitive
Quoted Post URL
2023-02-09 20:28:00+00:00
2024-07-07 18:08:28+00:00
Travel safely with your 100% free Concealed Carry Map guide and discover where you can and can't go with your firearm.
54653299
en
1318
4477
203
2002
29355033
2023-03-13 19:41:00+00:00
2024-07-06 19:19:48+00:00
Protect your money now! Two US banks collapsed, billions trapped. Invest in gold, a retirement hedge outside of the government's control.
1007312685413773312
en
292
388
39
320
23383507
2023-05-29 02:57:00+00:00
2023-12-28 21:11:50+00:00
Learn Coins and Their Value
1582662363530285056
en
286
380
17
285
31459875
2023-06-07 09:45:58+00:00
2023-12-28 20:56:11+00:00
Want to quickly expand your network? Bigo provides you with the best platform
1615603634448011265
en
396
276
21
494
214972443
2023-07-04
2023-12-28
Happy Birthday America!
1636590253
en
798
1086
88
44
1480370


iLEAPP 2026.4.5-dev
Dark Switch
SAVED REPORTS
Report Home
TWITTER X
Twitter X - Cached Posts
Twitter X - Cached Users
Twitter X - Notifications
Cache Updated
Username
User ID
Bio
Followers
Following
Favourites
Blue Verified
Protected
Profile Image URL
2023-12-28 18:01:12+00:00
Apple
380749300
https://t.co/4dZx1rwvgY
0
Yes
https://pbs.twimg.com/profile_images/1717013664954499
2023-12-28 18:01:12+00:00
tim_cook
1636590253
Apple CEO || Auburn Duke National Parks "Life's most persistent and urgent question is, 'What are you doing for others?'" - MLK. he/him
2039
Yes
https://pbs.twimg.com/profile_images/1535420431766671
2023-12-28 18:01:12+00:00
SemradF64851
1727967682094628864
32
Yes
https://pbs.twimg.com/profile_images/1727967745294364
2023-12-28 18:01:12+00:00
gelvin93847
1736620623663964160
0
Yes
https://pbs.twimg.com/profile_images/1736620688151296
2023-12-28 18:01:12+00:00
mattiaep
1322776140
CEO and Digital Forensics Analyst @ REALITY NET (https://t.co/G04RoVyJWU) Researcher at IGSG-CNR (https://t.co/ASkXLx1Gvk) SANS Certified Instructor (https://t.co/GMLheLo8nh)
1565
https://pbs.twimg.com/profile_images/1478811493114290
2023-12-28 18:01:12+00:00
rj_chap
18293321
DFIR & malware analyst. @sansforensics FOR528 Author & FOR610 Instructor. @CactusCon crew. Husband & father. Comments = own.
9447
Yes
https://pbs.twimg.com/profile_images/1710688349785313
2023-12-28 18:01:12+00:00
gregjoz
432943643
Apple SVP Marketing || Big believer in hw, sw and services integration @UMich Animal lover Biking & the great outdoors
393
Yes
https://pbs.twimg.com/profile_images/1535411147087790


iLEAPP 2026.4.5-dev
Dark Switch
SAVED REPORTS
Report Home
TWITTER X
Twitter X - Cached Posts
Twitter X - Cached Users
Twitter X - Notifications
725479270958694/database/modelCache.sqlite3
Show 15 entries
Search:
Timestamp
Text
Icon
From Users
Notification ID
2023-11-29 01:26:03+00:00
It's your X anniversary! Celebrate with a special post created just for you
milestone_icon
DtMcCcBXcAEAAAABSQ3bEbGWbBrp5tbMnfo
2023-12-28 18:01:10+00:00
There was a login to your account @TDfir from a new device on Dec 28, 2023. Review it now.
bird_icon
DtMcCcBXcAEAAAABSQ3bEbGWbBopXdLK7ol
2024-07-06 20:01:34+00:00
ldehner505 followed you
person_icon
DtMcCcBXcAEAAAABSQ3bEYI89UC2vZKHdOk
2024-07-06 20:01:34+00:00
ldehner505 followed you
person_icon
DtMcCcBXcAEAAAABSQ3bEY68hpQOYhyrdOk
2024-07-06 20:01:34+00:00
ldehner505 followed you
person_icon
DtMcCcBXcAEAAAABSQ3bEZP1Ch0OYhyrdOk
2024-07-06 20:01:34+00:00
ldehner505 followed you
person_icon
DtMcCcBXcAEAAAABSQ3bEZnVjg4OYhyrdOk
2024-07-06 20:01:34+00:00
ldehner505 followed you
person_icon
DtMcCcBXcAEAAAABSQ3bEZ8Qo6lOYhyrdOk
2024-07-08 10:46:00+00:00
Yakumono
recommendation_icon
DtMcCcBXcAEAAAABSQ3bEY68hpQbRvKakE4
2024-07-08 10:46:00+00:00
Yakumono
recommendation_icon
DtMcCcBXcAEAAAABSQ3bEZP1Ch0bRvKakE4
2024-07-08 10:46:00+00:00
Yakumono
recommendation_icon
DtMcCcBXcAEAAAABSQ3bEZnVjg4bRvKakE4
2024-07-08 10:46:00+00:00
Yakumono
recommendation_icon
DtMcCcBXcAEAAAABSQ3bEZ8Qo6lbRvKakE4
2024-07-08
Maggie Haberman
recommendation_icon
DtMcCcBXcAEAAAABSQ3bEY68hpSd5CHWkE4

## Ex.No : 05 IMPLEMENTATION OF DATA EXTRACTION FROM AN ANDROID DEVICE USING ADB

**Date :**

### AIM:

To extract a file from an Android emulator using Android Debug Bridge (ADB) and verify the integrity of the extracted file using the SHA-256 hashing algorithm.

### SOFTWARE?EQUIPMENTS REQUIRED:

- Windows 10/11 or Linux (Ubuntu/Kali)
- Android Studio software.
- Android Debug Bridge (ADB) platform tools.

### THEORY:

Android Debug Bridge (ADB) is a command-line tool that enables communication between a computer and an Android device or emulator. It consists of three components: the ADB client, ADB server, and ADB daemon (adbd). In digital forensics, ADB is used to retrieve device information, execute permitted shell commands, and transfer files for examination. The adb devices command identifies connected devices, adb push transfers files from the computer to the Android device, and adb pull extracts files from the device to the computer. In this experiment, an Android emulator is used to demonstrate file transfer and extraction. The SHA-256 cryptographic hash algorithm is then applied to the original and extracted files to compare their hash values and verify file integrity. Matching hash values indicate that the file contents are identical with respect to the SHA-256 comparison. Proper documentation, authorization, and preservation of original evidence are essential for maintaining forensic reliability.

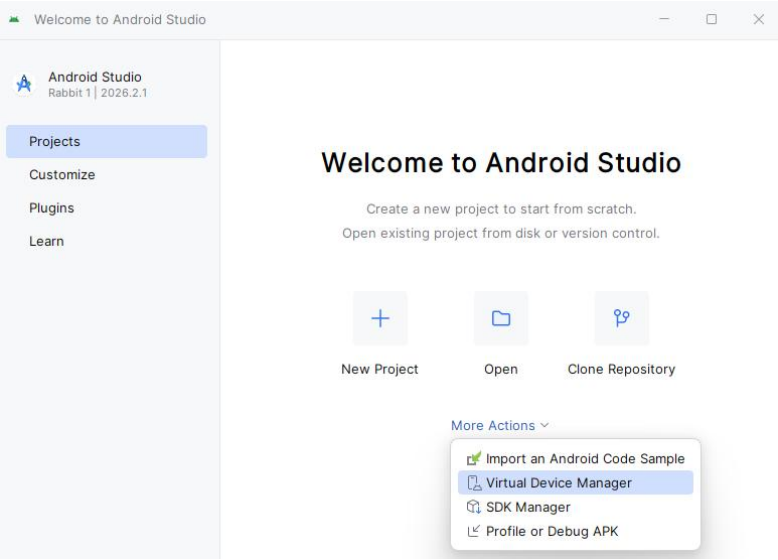
### ALGORITHM:

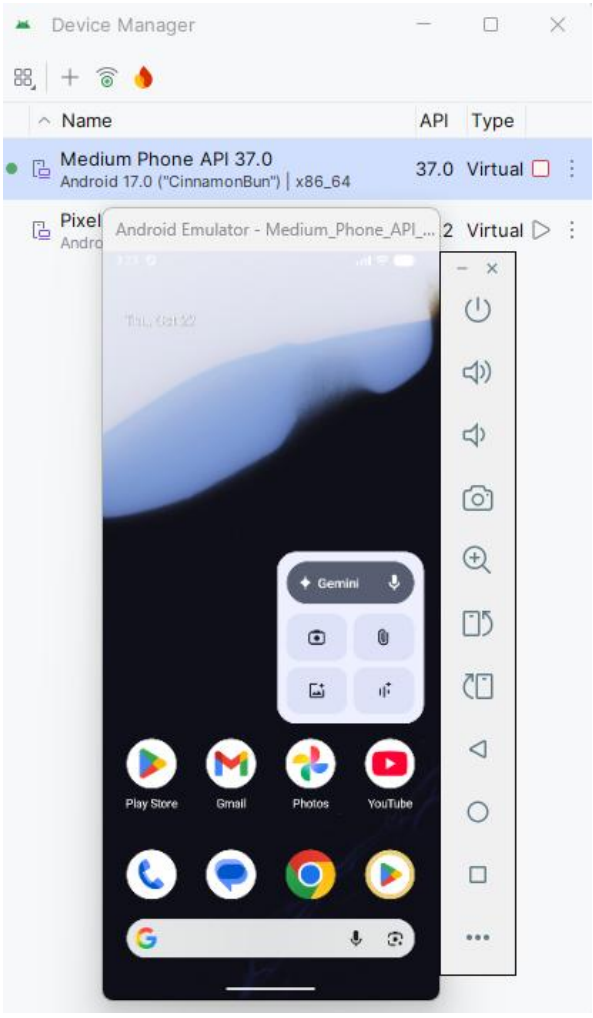
1. Download and install ExifTool.
2. Start the Android emulator using Android Studio.
3. Verify the ADB installation and emulator connection.
4. Retrieve the Android device information using ADB commands.
5. Create a sample text file on the computer.
6. Transfer the sample file to the Android emulator using the `adb push` command.
7. Verify the presence of the file on the emulator.
8. Extract the file from the emulator to the computer using the `adb pull` command.
9. Calculate the SHA-256 hashes of the original and extracted files.
10. Compare both hash values to verify file integrity.
11. End

## PROCEDURE

### Step 1: Start the Android Emulator

1. Open Android Studio.
2. Navigate to Tools → Device Manager.
3. Create a virtual Android device if one is not available.
4. Start the emulator.



**Step 2: Verify ADB Installation**

Open Command Prompt and execute:

adb version

This command displays the installed ADB version.

Change directory if needed: cd C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools

**Step 3: Check Device Connection**

Execute: adb devices

Verify that the emulator appears in the device list with the status device.

**Step 4: Collect Device Information**

Execute the following commands:

adb shell getprop ro.build.version.release

adb shell getprop ro.product.model

adb shell getprop ro.product.manufacturer

These commands retrieve the Android version, device model, and manufacturer.

```
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhamu>cd C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb devices
List of devices attached
emulator-5554   device


C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.build.version.release
17

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.product.model
sdk_gphone64_x86_64

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.product.manufacturer
Google

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>
```

**Step 5: Create a Sample File**

Execute:

mkdir "%USERPROFILE%\Documents\MobileForensicsLab"

echo Digital Forensics Lab Sample >

"%USERPROFILE%\Documents\MobileForensicsLab\sample.txt"

This creates a sample text file on the computer.

**Step 6: Transfer the File to the Emulator**

Execute:

adb push "%USERPROFILE%\Documents\MobileForensicsLab\sample.txt" /sdcard/Download/sample.txt

The file is copied from the computer to the emulator.

**Step 7: Verify the File on the Emulator**

Execute:

adb shell ls -l /sdcard/Download/sample.txt

adb shell cat /sdcard/Download/sample.txt

These commands verify the file's presence and display its contents.

**Step 8: Extract the File from the Emulator**

Execute:

adb pull /sdcard/Download/sample.txt "%USERPROFILE%\Documents\MobileForensicsLab\extracted_sample.txt"

The file is copied back from the emulator to the computer.

**Step 9: Verify File Integrity**

Calculate the SHA-256 hash of the original file:

certutil -hashfile "%USERPROFILE%\Documents\MobileForensicsLab\sample.txt" SHA256

Calculate the SHA-256 hash of the extracted file:

certutil -hashfile "%USERPROFILE%\Documents\MobileForensicsLab\extracted_sample.txt" SHA256

Compare the two hash values. If both values match, the original and extracted files have identical contents with respect to the SHA-256 comparison.

**OUTPUT:**

SHA-256 hash of the original file:

SHA-256 hash of the extracted file: